```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

url = "https://raw.githubusercontent.com/shivamb/netflix-shows/master/netflix_titles.csv"

try:
    df = pd.read_csv(url)
except Exception:
    from google.colab import files
    uploaded = files.upload()
    df = pd.read_csv(next(iter(uploaded)))

print("First five records:")
display(df.head())

print("Last five records:")
display(df.tail())

print("Shape:", df.shape)

df.info()

print("\nMissing values:")
print(df.isnull().sum())

print("\nSummary statistics:")
display(df.describe(include="all").T)
```

Choose Files netflix_titles.csv

netflix_titles.csv(text/csv) - 3399671 bytes, last modified: 10/12/2026 - 100% done

```
Saving netflix_titles.csv to netflix_titles (1).csv
First five records:
```

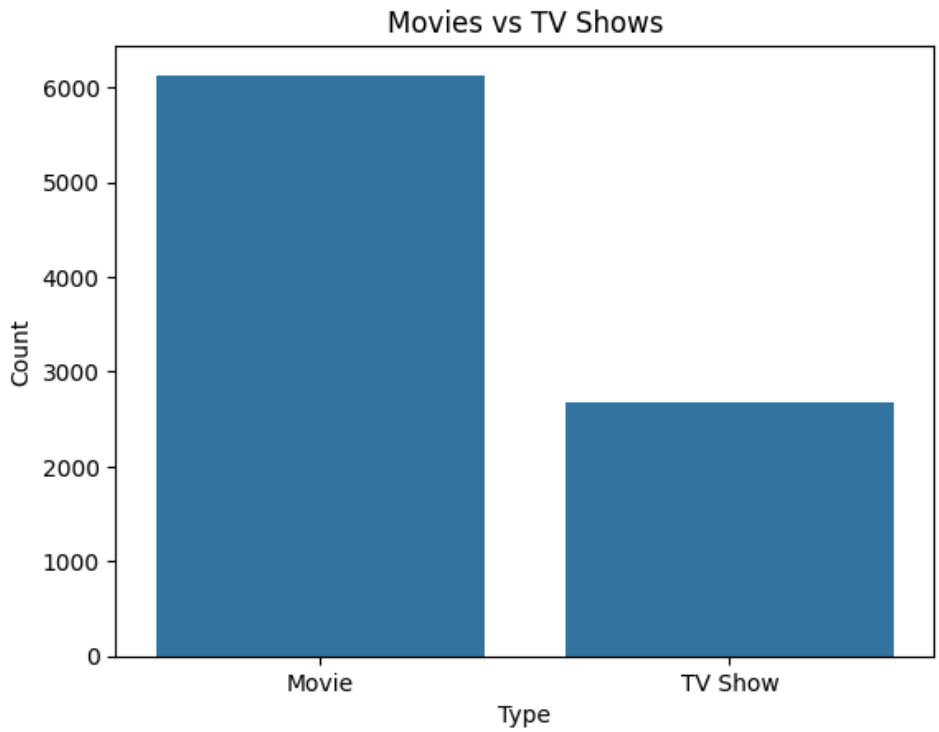
	show_id	type	title	director	cast	country	date_added	release_year	rating	durat
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	Seas
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Sea
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Sea
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	Seas

```
Last five records:
```

	show_id	type	title	director	cast	country	date_added	release_year	rating
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG
8806	s8807	Movie	Zubaan	Mozez	Vicky Kaushal, Sarah-	India	March 2,	2015	TV-14

```
sns.countplot(data=df, x="type")

plt.title("Movies vs TV Shows")
plt.xlabel("Type")
plt.ylabel("Count")
plt.show()
```



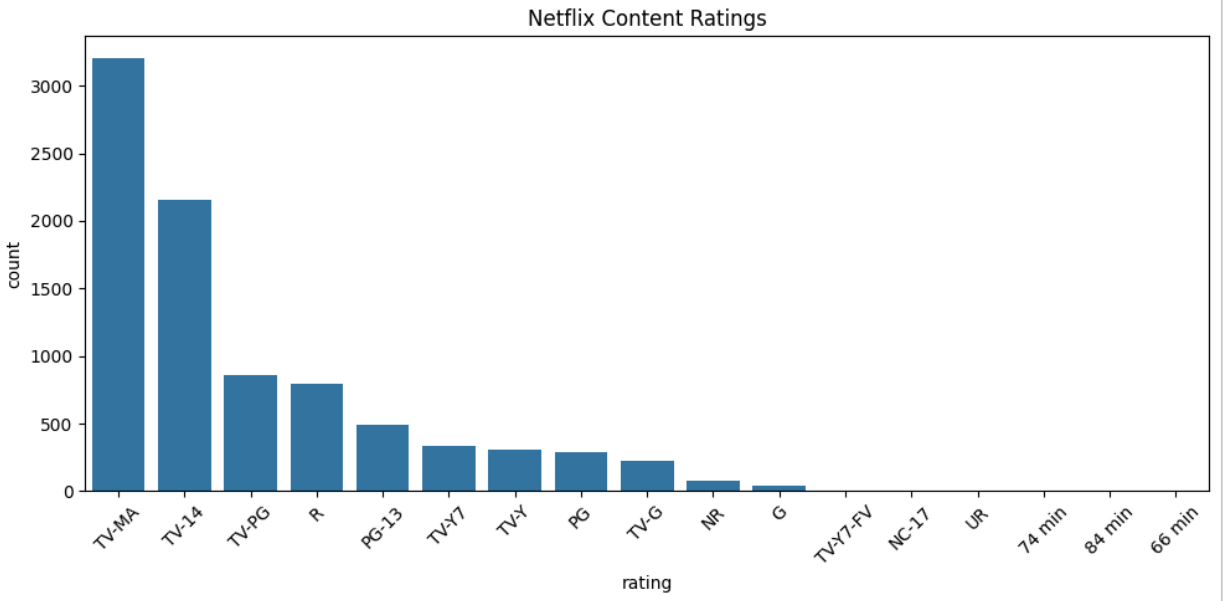
listed_in 0

```
plt.figure(figsize=(10, 5))

sns.countplot(
    data=df,
    x="rating",
    order=df["rating"].value_counts().index
)

plt.title("Netflix Content Ratings")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

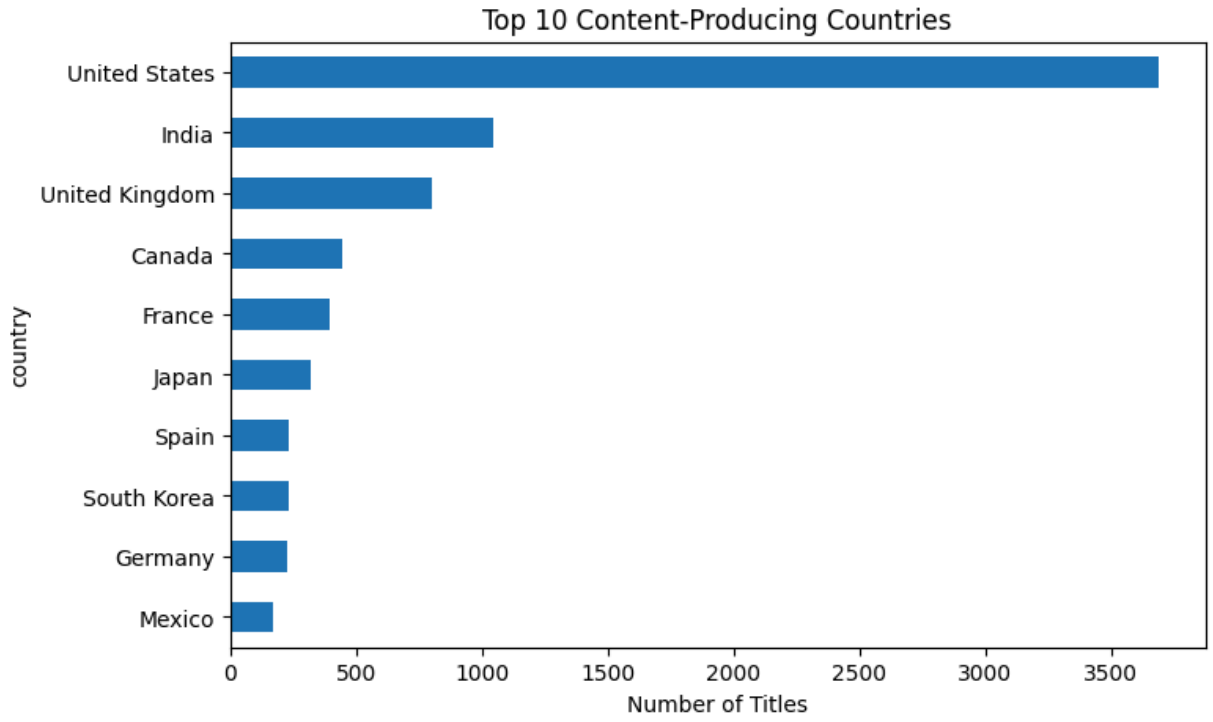
cast 7982 7692 19 NaN NaN NaN NaN NaN NaN



```
countries = (
    df["country"]
    .dropna()
    .str.split(", ")
    .explode()
    .value_counts()
    .head(10)
)

countries.sort_values().plot(kind="barh", figsize=(8, 5))

plt.title("Top 10 Content-Producing Countries")
plt.xlabel("Number of Titles")
plt.show()
```




```
df["date_added"] = pd.to_datetime(
    df["date_added"].astype(str).str.strip(),
    errors="coerce"
)

df["date_added_year"] = df["date_added"].dt.year

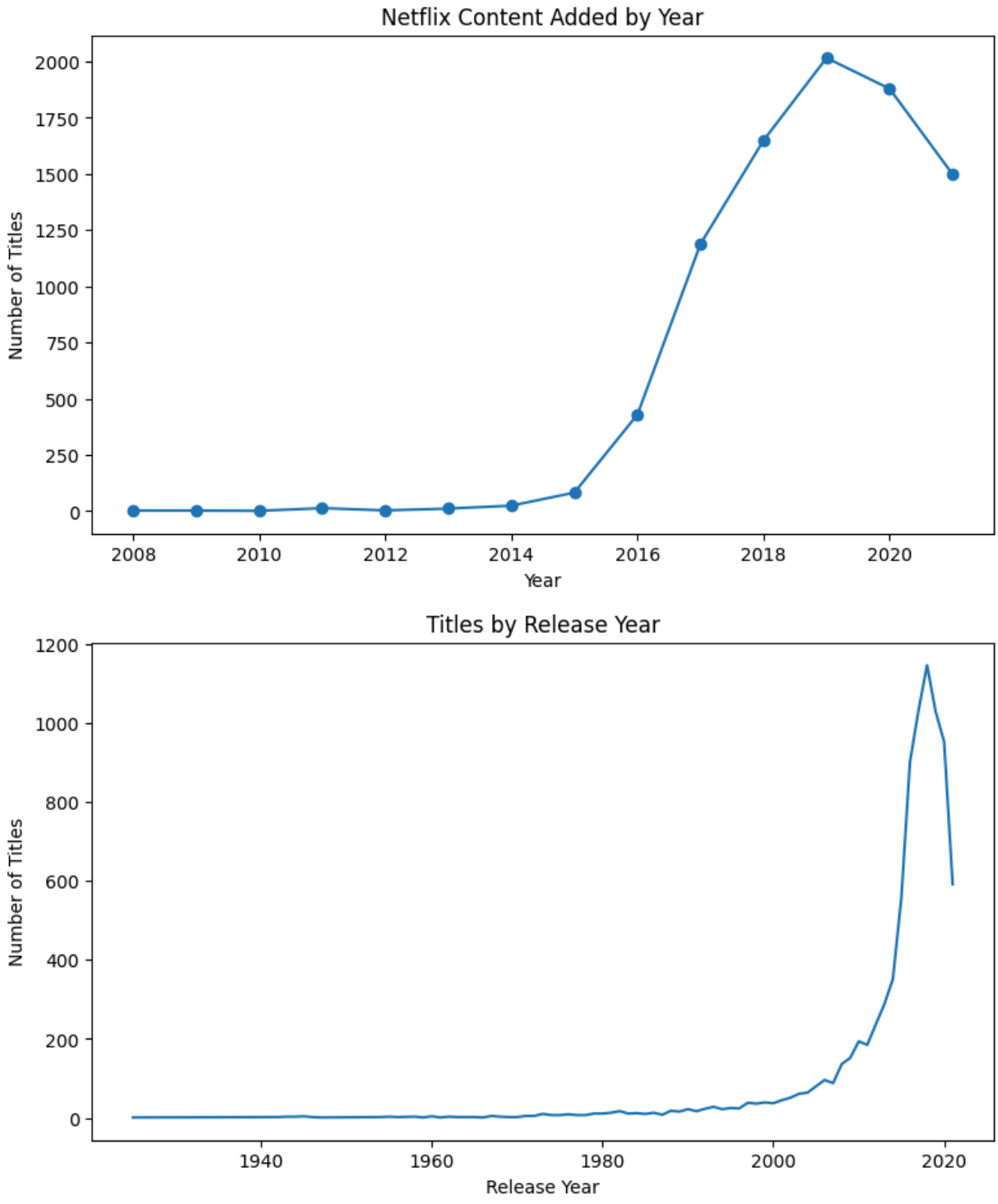
df["date_added_year"].value_counts().sort_index().plot(
    kind="line",
    marker="o",
    figsize=(9, 5)
)

plt.title("Netflix Content Added by Year")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.show()

df["release_year"].value_counts().sort_index().plot(
    kind="line",
    figsize=(9, 5)
)

plt.title("Titles by Release Year")
```

```
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```




```
df["duration_num"] = pd.to_numeric(
    df["duration"].astype(str).str.extract(r"(\d+)")[0],
    errors="coerce"
)

movies = df[df["type"] == "Movie"]
tv_shows = df[df["type"] == "TV Show"]

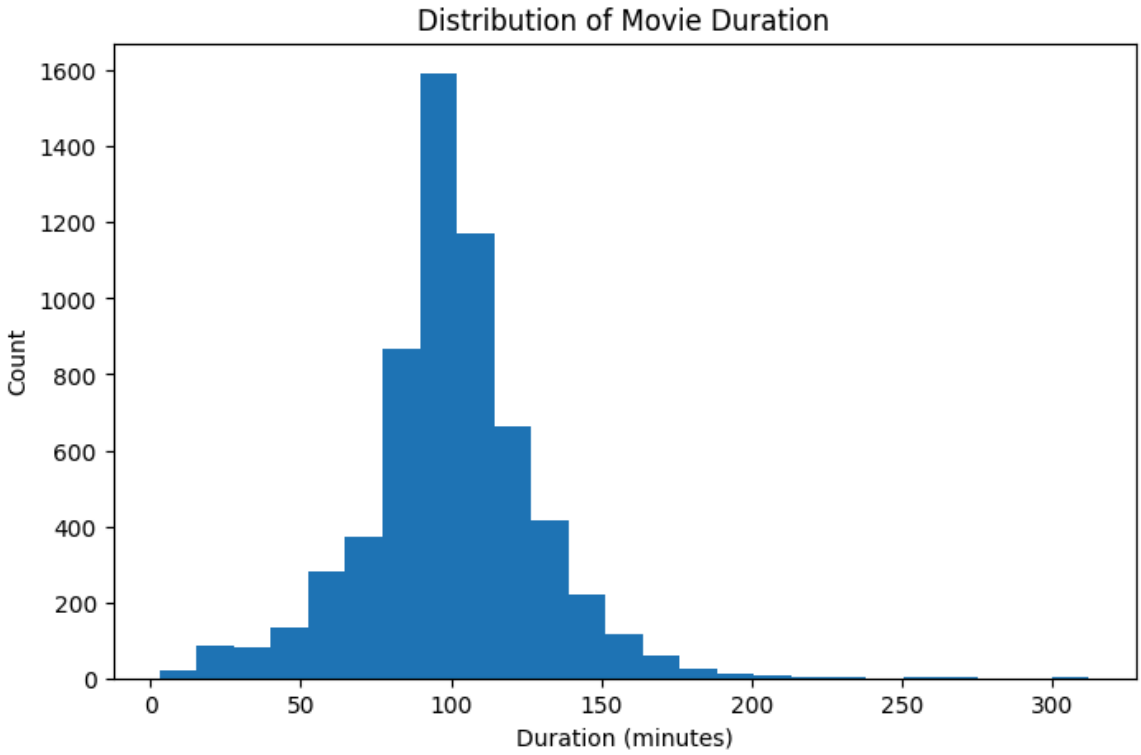
plt.figure(figsize=(8, 5))
plt.hist(movies["duration_num"].dropna(), bins=25)
```

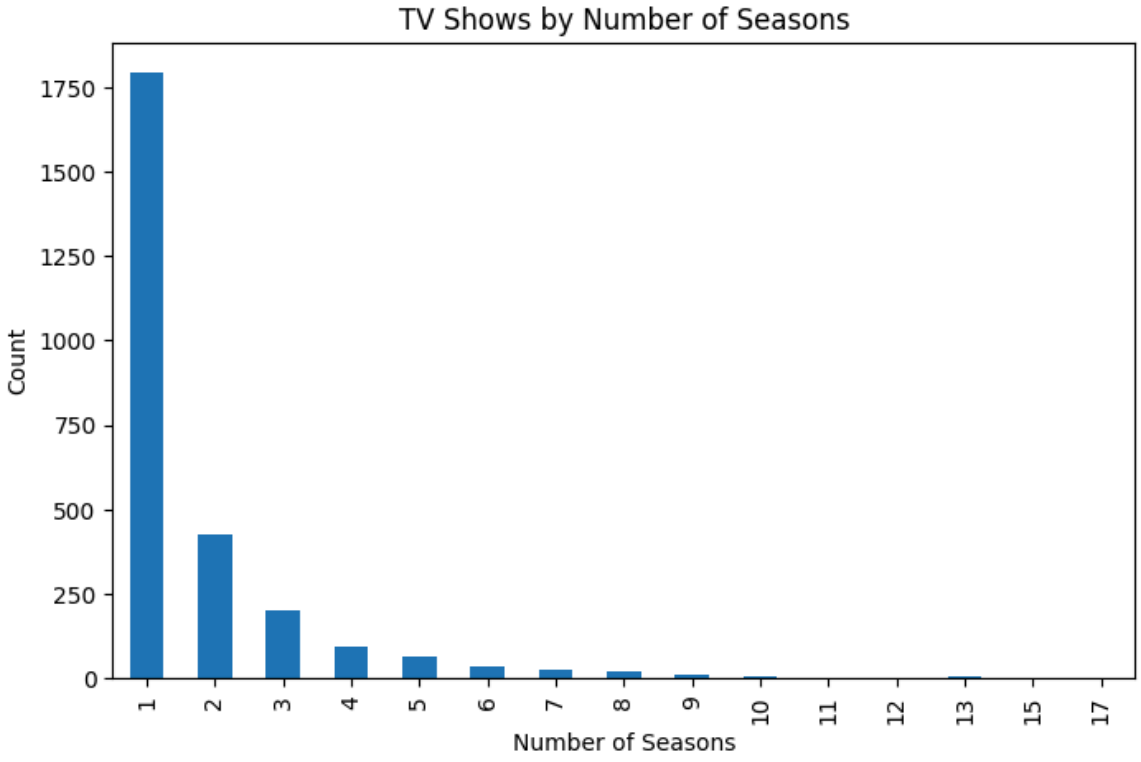
```
plt.title("Distribution of Movie Duration")
plt.xlabel("Duration (minutes)")
plt.ylabel("Count")
plt.show()

seasons = pd.to_numeric(
    tv_shows["duration"].astype(str).str.extract(r"(\d+)")[0],
    errors="coerce"
)

seasons.value_counts().sort_index().plot(
    kind="bar",
    figsize=(8, 5)
)

plt.title("TV Shows by Number of Seasons")
plt.xlabel("Number of Seasons")
plt.ylabel("Count")
plt.show()
```






```
genres = (
    df["listed_in"]
    .dropna()
    .str.split(", ")
    .explode()
    .value_counts()
)

print("Top 10 genres:")
display(genres.head(10))

print("Top 10 directors:")
display(
    df["director"]
    .dropna()
```

```
        .str.split(", ")
        .explode()
        .value_counts()
        .head(10)
    )

    print("Top 10 actors:")
    display(
        df["cast"]
        .dropna()
        .str.split(", ")
        .explode()
        .value_counts()
        .head(10)
    )
```

```
Top 10 genres:
```

```
!pip -q install wordcloud

from wordcloud import WordCloud

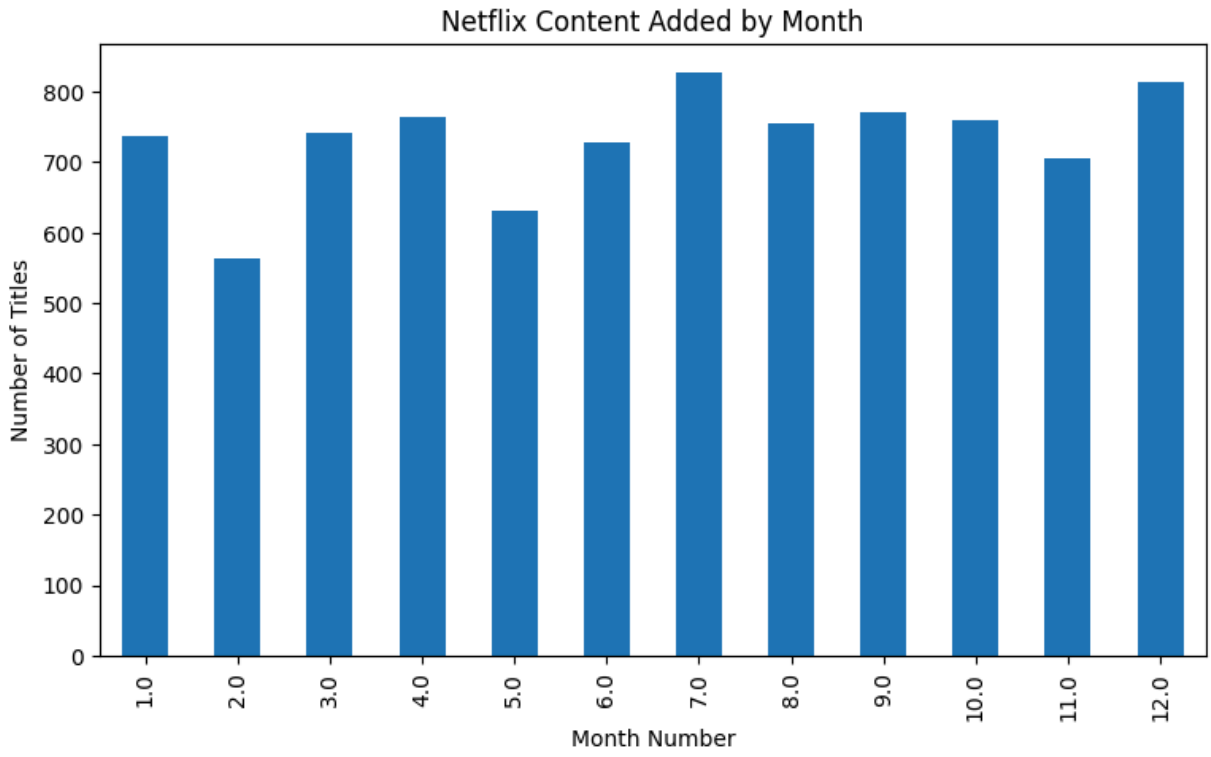
text = " ".join(df["description"].dropna().astype(str))

wordcloud = WordCloud(
    width=1200,
    height=600,
    background_color="white"
).generate(text)

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Common Words in Netflix Descriptions")
plt.show()
```

Romantic Movies 616


Common Words in Netflix Descriptions
world
life
find
friend
love
family
help
take
become
new
woman
save
team
must
make
live
young
two
one
fall
way
turn
daughter
series
secret
group
three
set
year
death
boy
girl
father
mother
murder
back
begin
time
meet
stand
power
struggle
discover
couple
battle
home
son
four
face
teen
plan
documentary
wife
husband
follow
relationship
journey
romance
navigate
challenge
wealthy
show
past
evil
event
business
career
come
stories
comedian
mysterious
city
share
embark
dream
school
history
comedy
detective
writer
best friend
father
work
travel
soon
begin
crime
fight
explore
around
girlfriend
brother
high school
friendship
marriage
seek
goes
leave
keep
head
sister
college
women
town
criminal
New York
India
village
behind
children
will
revenge
music
American
start
quest
gang
gets
star
turn
end
uncover
cop
join
go
big
year old
lead
parent
crew
look
together
win
pal
force
forced
stage
learn
rise
whose
use
navigate
wealthy
challenge
show
past
kid
evil
event
business
human
girl
young
woman
career
come
stories
tries
return
people
first
struggling
bring
Based
personal
boy
film
now
night
death
old
mysterious
hit
land
comedian
share
city
living
embark
dream
school
history
comedy
detective
writer
best friend
father
young man
long
put
change
story
play
determined
dark
bond
child
former
murder
stand
power
struggle
day
back
move
dangerous
mission
meet
team
must
make
live
young
two
one
fall
way
wife
heart
form
try
search
stop
five
soon
travel
work
comic
may
animal
investigate
thing
time
crime
fight
local
Earth
explore
powerful
different
ex
girlfriend
around
group
three
discover
couple
battle




```
Month with most content added: 7.0
```

```
country_df = df.dropna(subset=["country"]).copy()

country_df["country"] = country_df["country"].str.split(", ")
country_df = country_df.explode("country")

country_type = pd.crosstab(
    country_df["country"],
    country_df["type"]
)

display(
    country_type.sort_values(
        by="Movie",
        ascending=False
    ).head(10)
)
```

type	Movie	TV Show
country		
United States	2751	938
India	962	84
United Kingdom	532	272
Canada	319	126
France	303	90
Germany	182	44
Spain	171	61
Japan	119	199
China	114	48
Mexico	111	58